# Transition of Care AI-Assisted (ToC-AI) Proof of Concept Leveraging the Clinical Health AI-augmented Records Toolkit (CHART)

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## Introduction & Objectives

### Clinical problem

Post-acute transitions are often hindered by incomplete, fragmented, and inconsistently represented handoff information. Clinical data are scattered across EHR fields, notes, and documents, resulting in duplication, conflicting information, unclear data provenance, and missing context. Receiving teams must often reconstruct the patient's clinical story with inadequate information.

### Why standards alone are insufficient

PACIO and FHIR enable standardized data exchange but do not inherently support clinical interpretation. They do not filter for relevance, reconcile conflicting or duplicate data, assess completeness or accuracy, prioritize actionable information, or provide the person-centric view needed for effective care transitions.

### Objective

To address these gaps, we demonstrate **ToC-AI**, a standards-led, AI-augmented toolkit that consolidates structured and unstructured data and aligns outputs with the PACIO Transitions of Care Implementation Guide. ToC-AI organizes and identifies relevant data elements, while clinicians retain review and reconciliation authority with visibility into source evidence. The goal is to transform exchanged data into complete, meaningful, timely, and transition-ready information for the next care team.

## Materials & Methods

### Prototype setting and systems

Synthetic post-acute patient scenarios combining structured and free-text clinical records were used to develop and evaluate the prototype. Development incorporated input from the PACIO Transitions of Care (ToC) Subgroup and was guided by FHIR, USCDI v5, and the PIQI Framework. The prototype consisted of two complementary systems:

- CHART (Clinical Health AI-augmented Records Toolkit), a reusable modular AI toolkit for clinical information extraction, analysis, synthesis, and transformation.

- ToC-AI (Transitions of Care AI-Assisted) Application, a proof-of-concept implementation that used CHART capabilities to orchestrate end-to-end transitions of care.

CHART is a standards-led, AI-augmented toolkit of services that consolidate structured and unstructured data into standardized and harmonized healthcare information. The ToC-AI App leverages CHART services to organize transition-relevant information while preserving clinician review, reconciliation, and source traceability, with the goal of producing complete, meaningful, and timely information for the next care team.

### AI-assisted data processing workflow

- Queried and ingested structured and unstructured EHR data.
- Extracted clinically relevant entities mapped to USCDI v5 elements.
- Applied element-level governance to support transition-level information generation.
- Normalized to standard terminologies as applicable.
- Detected duplicate and conflicting information.
- Presented flagged information for clinician review and reconciliation.
- Packaged reconciled content into a FHIR ToC-aligned document and bundle.

### Clinician review and reconciliation

- Clinicians reviewed AI-suggested content.
- Assessed clinical relevance, source support, and appropriateness for inclusion.
- Resolved conflicting or inconsistent information.
- Determined readiness for downstream sharing.

AI-identified data issues, source evidence, and recommendations supported human review before transition information was finalized for sharing.

### Standards and quality approach

Standards and quality controls were applied throughout the ToC-AI workflow. The Analysis Agent used USCDI v5 and Terminology MCP services to identify and normalize clinical data using SNOMED CT, LOINC, RxNorm, and ICD-10-CM. The Data Synthesis Agent reconciled conflicting information and applied confidence scoring, while Patient Information Quality Improvement (PIQI) informed validation and scorecards enabled data-quality assessment. The Data Conversion Agent, supported by the PIQI-to-FHIR MCP service, transformed reconciled content into PACIO ToC-conformant FHIR bundles. A separate Quality Evaluation Agent assessed extraction quality, PIQI scorecards, and ToC FHIR conformance.

## Findings and Discussion

### Functional proof-of-concept findings

ToC-AI extracted structured and free-text data from synthetic scenarios, identified and mapped key USCDI v5 elements, detected duplicate and conflicting information, and produced quality-scored,

clinician-reviewable transition data. The prototype also generated FHIR bundles aligned with the PACIO Transitions of Care (ToC) Implementation Guide (IG).

## Clinical workflow relevance

The prototype consolidated fragmented source data into a unified handoff view, surfaced key transition information and potential risks, and preserved provenance for clinician review. This approach may reduce manual chart-review burden, although that outcome was not measured in the proof of concept.

## Informatics insight

The exploration suggests the value of interoperability extends beyond data movement to semantic structuring, relevance filtering, and interpretability. By organizing standards-based data into clinically meaningful information, the approach bridges the gap between FHIR exchange and clinical usability.

## Evidence boundary

Findings reflect synthetic proof-of-concept scenarios only. No real-world deployment, clinical outcomes, or efficiency measures were assessed, and potential reductions in chart-review burden remain hypotheses for future testing. The prototype demonstrated technical feasibility rather than clinical effectiveness.

**Table 1. Evidence boundary: proof of concept vs. future exploration**

| Evaluation area | Demonstrated in proof of concept | To be measured in pilots |
|---|---|---|
| **Data ingestion & source handling** | Processed structured and unstructured clinical data from curated synthetic post-acute care scenarios. | Reliability across real EHR, HIE, FHIR, file-share, and document-repository inputs, including source variability and connector performance. |
| **Clinical data extraction** | Identified clinically relevant information from structured fields and free-text notes. | Extraction accuracy and completeness across representative post-acute care data types and documentation styles. |
| **Standards mapping** | Mapped candidate information to key USCDI v5 and PACIO Transitions of Care (ToC) elements. | Mapping accuracy and completeness against the PACIO ToC Implementation Guide across real-world cases. |
| **Duplicate & conflict reconciliation** | Identified duplicate and conflicting entries and generated reconciled candidate information for review. | Conflict-detection precision and clinician agreement with AI-generated reconciliation suggestions. |
| **Confidence & data-quality indicators** | Generated confidence-scored candidate information using PIQI-informed data-quality logic. | Quantitative validation of confidence scoring and quality-assurance methods across broader clinical datasets. |
| **Source traceability** | Preserved source evidence and manifests so AI-assembled content could be traced back to originating artifacts. | Clinician use and perceived usefulness of source evidence during review, including its contribution to efficient adjudication. |
| **Clinician review & human reconciliation** | Provided a clinician-reviewable interface in which users could review AI-generated content, resolve conflicts, and determine readiness for downstream sharing. | Clinician agreement, usability, reconciliation consistency, and workflow fit in realistic care settings. |

| Evaluation area | Demonstrated in proof of concept | To be measured in pilots |
|---|---|---|
| **FHIR ToC output & conformance** | Transformed harmonized data into clinician-reviewable FHIR bundles aligned with the PACIO ToC Implementation Guide. | Formal FHIR conformance validation across a broader set of cases and downstream systems. |
| **Review burden & efficiency** | Not quantitatively measured; the proof of concept demonstrated a workflow intended to reduce manual abstraction and chart review. | Time to review, manual abstraction effort, clinician burden, and performance relative to baseline workflows. |
| **Real-world workflow integration** | Defined an intended deployment architecture, but did not evaluate production EHR/HIE integration or operational deployment. | Usability, workflow integration, operational feasibility, security and governance, and controlled deployment in real post-acute care workflows. |
| **Clinical & transition outcomes** | Not measured; the proof of concept does not support claims of improved clinical outcomes or efficiency. | Transition quality and other clinically meaningful outcomes in real-world post-acute care settings. |

### Future exploration

Future research should evaluate extraction accuracy and completeness against PACIO ToC FHIR IG content expectations, conflict-detection precision, clinician agreement with AI suggestions, FHIR conformance, data quality, clinician review time, usability, and workflow integration.

## Conclusions

This proof of concept demonstrates the feasibility of applying Generative AI to structured and unstructured clinical data to produce reviewable, transition-relevant information. PACIO and FHIR provide the interoperability foundation, while AI enhances the usability of standards-based exchange through semantic structuring and relevance filtering. Human oversight remains essential for clinical review, reconciliation, safety, and trust.

These conclusions are limited to technical feasibility and do not establish improvements in clinical outcomes, efficiency, or real-world workflow performance. Next steps include security and governance review, real-world validation, clinician usability studies, controlled pilot deployment, and benchmarking against baseline workflows. The goal is not simply to exchange data, but to make interoperable data clinically meaningful, actionable, and ready for the next care team.

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## Contact & Call to Action

Collaborate with PACIO to evaluate and pilot AI-assisted, clinician-governed ToC workflows in real-world post-acute care settings.

Contact email: info@pacioproject.org

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